

Vaasu Sohee

Data Scientist

517-490-7865 · vsohee@gmail.com · [LinkedIn](#) · [Portfolio](#) · [ICMLANT Publication](#)

PROFESSIONAL EXPERIENCE

Data Scientist Intern (Co-op) · Delta Dental Insurance

Jun 2025 – Present

- Improved model recall **40%+ across 3M+ healthcare claims** by leading a self-supervised transformer embedding framework using PyTorch Lightning, contrastive pretraining, Optuna, and MLflow.
- Enabled NL analytics across **131 datasets** via Provider Insights AI (Snowflake + Azure OpenAI), benchmarking **4 LLMs** on a 20-question golden set (80 runs) for correctness/token usage with LLM-as-a-Judge and human feedback.
- Eliminated **hundreds of manual review hours** by building an OCR + GPT-5.5 mini document-intelligence pipeline with few-shot prompting and Pydantic validation to structure thousands of handwritten and non-standard invoices.
- Powered Medicare/Medicaid actuarial and government-program analytics on multimillion-row claims data by engineering partition-aware Oracle SQL and Dataiku pipelines with data-validation workflows.

Data Scientist · Ernst & Young

Jan 2023 – Jun 2024

- Reduced ML retraining cycles **30%** and improved inference accuracy **15%** using AWS SageMaker + Bedrock, SageMaker Feature Store, and automated retraining for Fortune 500 migrations.
- Cut ETL latency **88%+, from 4+ hours to under 30 minutes**, by re-engineering Python/SQL/Airflow/Spark pipelines on AWS S3/Athena processing 5M+ records/day.
- Delivered a **30% revenue KPI lift** across Fortune 100 portfolios by designing A/B tests and uplift trees with CausalML + SciPy and analyzing experiment results in Python, SQL, and Tableau.
- Accelerated ML deployment velocity **25%** with zero-downtime rollouts by containerizing FastAPI services via Docker and Git-based CI/CD across multi-region AWS environments.

Data Science Intern · Ernst & Young

Jun 2022 – Aug 2022

- Built and deployed cross-validated classification and regression models across 10+ client engagements, delivering a 20% improvement in submission rates and improving decision quality by 25% for Operations and Marketing teams.
- Applied uplift modeling and controlled experiments to provide advisory insights to client leadership, consulting on data-driven strategies for non-technical stakeholders.

Data Science Intern · NIIT

Jun 2021 – Dec 2021

- Developed a CNN-based deep learning model in PyTorch achieving 95%+ accuracy, deployed with real-time inference pipelines at sub-100ms latency and automated drift monitoring dashboards.

SELECTED PROJECTS

AgentGuard - Security gateway for AI agents · [Link](#) · Independent Project

May 2026 – Aug 2026

- Built an AI agent tool runtime gateway (MCP, Next.js, TypeScript) enforcing **5 runtime controls**: provenance checks, policy rules, secret stripping, human approval for risky actions, and tamperproof audit logging.
- Integrated OpenAI Agents SDK, LangGraph, Mastra, and PydanticAI agents with a shared backend via LiteLLM; stress tested against **4 realworld attacks**: prompt injection, secret leaks, unauthorized rollbacks, and unauthorized access.
- Built a 3part policy intelligence layer: semantic search (Qdrant/LlamaIndex), audit history (PostgreSQL), and classifier tuning/tracing (DSPy, Arize Phoenix).

Demand & Capacity Forecasting Capstone · (Signed NDA) · Jackson National Life

Aug 2025 – Dec 2025

- Built a multi segment forecasting pipeline using Temporal Fusion Transformer and SARIMAX with exogenous regressors across 40+ contract segments, reducing RMSE by **22%** and MAPE by **18%** versus baseline.
- Translated model outputs into **\$3.5M in annual risk reduction** through optimized hedging strategies, crediting rate guarantees, and RILA pricing decisions in collaboration with business stakeholders.

[Portfolio](#) - More projects

EDUCATION

M.S. Data Science · Michigan State University · GPA: 3.92 / 4.0

Aug 2024 – May 2026

B.Tech Computer Science · NIIT University · GPA: 3.6 / 4.0

Aug 2019 – May 2023

SKILLS

Cloud & Data Infrastructure: AWS (S3, SageMaker, Athena, Bedrock), Snowflake, Docker

Data Engineering & Pipelines: ETL/ELT pipeline design, Apache Airflow, Apache Spark, Dataiku

Programming: Python (Pandas, NumPy, scikit-learn), SQL (Oracle, PostgreSQL)

ML & MLOps: PyTorch, MLflow, FastAPI, CI/CD, LangChain, LangGraph, RAG, Prompt Engineering, Pydantic validation

LLM Evaluation & Observability: LLM-as-a-Judge, LangSmith, Azure OpenAI

Analytics & Modeling: Demand forecasting, A/B testing, Causal inference, Predictive analytics